
PROJECT PROPOSAL

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PROJECT PROPOSAL

TITLE: PRECLINICAL ASSESSMENT OF ANTI-UROLITHIATIC ACTIVITY OF *Moringa oleifera* pod wall extract IN EXPERIMENTAL ANIMALS

AIM:

To evaluate the anti-urolithiatic activity of the ethanolic extract of *Moringa oleifera* pod wall in ethylene glycol– and yeast extract–induced urolithiasis in Wister rats.

OBJECTIVES:

To evaluate the parameters like

- ✓Induction of disease.
- ✓Effect of test drug on urinary parameters
- ✓Identification of biomarkers (Calcium, Magnesium).
- ✓Comparing results with the standard drug

INTRODUCTION:

The term urolithiasis refers to the formation of stones anywhere within the urinary tract, including the kidneys, ureters, bladder, or urethra. This multifactorial disorder arises when minerals such as calcium, phosphate, and oxalate accumulate and crystallize in supersaturated urine. As these microscopic crystals grow and cluster together, they can form larger stones that may travel or become lodged within the urinary passages, leading to a range of clinical issues. Urolithiasis is clinically significant because of its widespread occurrence, tendency to recur, and its ability to cause notable illness and strain on healthcare systems. It is a frequent reason for emergency medical visits, commonly presenting with severe flank pain, hematuria, nausea, and vomiting. When stones obstruct urine flow or irritate the urinary tract, they may lead to repeated urinary tract infections. If left untreated, urolithiasis can result in serious complications such as urinary obstruction, hydronephrosis, progressive kidney damage, and in severe cases, life-threatening infections or loss of renal function. The recurrent nature of the condition often diminishes patients' quality of life and requires repeated medical or surgical interventions. Therefore, early detection, appropriate treatment, and preventive strategies are essential to minimize complications, reduce recurrence, and lower the long-term healthcare burden associated with the disease.

HISTOPATHOLOGY ASSESSMENT:

1. Sacrifice and Tissue Collection:

- At the end of the experiment, rats were sacrificed.
- Both kidneys were carefully excised and rinsed with saline to remove blood and debris.

2. Fixation:

- Kidneys were fixed in **10% neutral buffered formalin** to preserve tissue structure.

3. Tissue Processing:

- Fixed tissues were processed through **standard histological procedures**.
- Tissues were embedded in **paraffin wax** for sectioning.

4. Sectioning:

- paraffin-embedded tissues were cut into **5 µm thick sections** using a microtome.

5. Staining:

- Sections were stained with **Hematoxylin and Eosin (H&E)** to visualize cellular and structural details.

6. Microscopic evaluation:

- Sections were examined under a **light microscope**.
- Evaluated parameters included:

- Renal tubular damage
- Interstitial inflammation
- Crystal deposition
- General structural alterations

7. Comparative analysis

- Histopathological findings were compared across all experimental groups.
- The analysis assessed the **protective effect of the ethanolic extract** against ethylene glycol-induced renal injury.

1. Types of Kidney Stones

Kidney stones are classified according to their chemical makeup, and each type, has a distinct origin and clinical significance:

1. Calcium Stones

This is the most common category of kidney stones, making up the majority of cases. These stones primarily include calcium oxalate and calcium phosphate. Excess levels of calcium or oxalate in the urine, often related to dietary or metabolic imbalances, promote their formation.

2. Uric Acid Stones

Uric acid stones occur when the urine is persistently acidic, allowing uric acid to crystallize. These are often associated with diets high in purines (found in red meat and certain seafood), increased cell turnover, or metabolic conditions like gout.

3. Struvite Stones

These stones are formed of magnesium ammonium phosphate and are typically linked to urinary tract infections caused by bacteria that produce urease. Struvite stones can grow quickly and may form large, branching (staghorn) stonee

4. Cystine Stones

Cystine stones are rare and result from a hereditary condition called cystinuria, in which the kidneys excrete excessive amounts of the amino acid cystine, leading to stone formation.

PLANT PROFILE:

SCIENTIFIC NAME : *Moringa oleifera* **pod wall extract**

FAMILY: Moringaceae

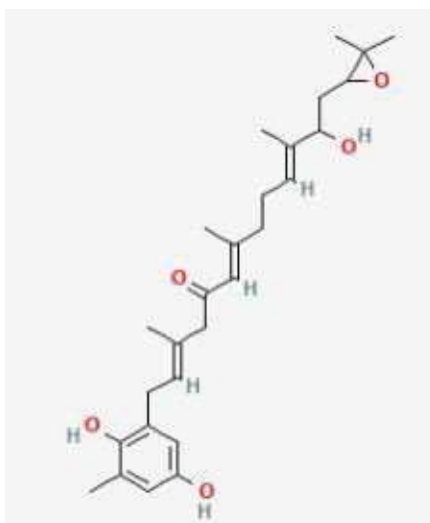
KINGDOM : Plantae

ORIGIN *Moringa oleifera* is **native to the sub-Himalayan regions of Northern India** and is widely cultivated throughout **India, Africa, Southeast Asia, and tropical regions worldwide** due to its medicinal and nutritional value.



DRUG PROFILE:

Class: Anti-urolithiatic agent



ANIMALS:

Adult male wistar albino rats weighing 150-250g are taken. They are divided into 5 groups. Each group contains 6 animals. The animals will be allocated in groups and allowed to accommodate for one week. They will be provided with a standard pellet diet and were given with water. The animals were kept at a temperature of 22 celusis.

And a 12 hours' light-dark cycles. All experiments were handled in accordance with the regulations of animal ethical committee.

EXPERIMENTAL DESIGN:

30 rats divided into five (n=6) groups, as follows Group-1: Normal group
Group-2: Diseased control (ethylene glycol) Group-3: Standard group (Cystone)
Group-4 : low dose(***Moringa oleifera* pod wall extract**)
Group-5: high dose (***Moringa oleifera* pod wall extract**)

It might take around 21 days to complete the work and induction is continued throughout the study period along with treatment.

EXPECTED RESULTS:

be *Moringa oleifera* herbal medicine will reduces or disappers the kidney stones in experimental animals that its action may **be *Moringa oleifera* pod wall extract** is expected to show dose-dependent anti-urolithiatic activity with significant preventive and curative effects.

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